| Method family | weightprog support | Validated on | Used in the companion study's runs (key below) |
|---|---|---|---|
| **Find the behaviour** | | | |
| Linear probes, read-outs [14] | ✓ yes | ○ tiny models only | • all 9 runs |
| Logit lens, direct logit attribution [15, 16] | ✓ yes | ○ tiny models only | • Fable r2, Astra r2, Opus inf |
| Activation capture at decision tokens [10–12] | ✓ yes | ● **real E4B** | • all 9 runs |
| Difference of means [17] | ✓ yes | ○ tiny models only | • recipe r1, Opus r1 |
| Exact prediction of an edit's effect (§2) | ✓ yes | ● **real E4B** | • Fable r2, Opus inf, a Fable 5.2 run |
| Activation patching [3, 18] | ✓ yes | ○ tiny models only | ○ none |
| Attention and head analysis [19] | ~ part | ○ tiny models only | • Fable r1 |
| Sparse autoencoders [20] | ✗ no | — | ○ none |
| **Change it** | | | |
| Gated logit-direction edit (§2; cf. [21]) | ✓ yes | ○ tiny models only | • Fable r2, Astra r2, Fable 5.2 rep |
| Unit (neuron) patching [21, 22] | ✓ yes | ● **real E4B** | • Opus r2, Opus inf, Fable 5.2 inf |
| Key → value update with covariance [3, 4] | ✓ yes | ○ tiny models only | ○ none |
| Null-space restriction [5] | ✓ yes | ○ tiny models only | ○ none |
| Directional ablation [23] | ✓ yes | ○ tiny models only | • recipe r1, Fable 5.2 rep |
| Steering written into weights [24] | ✓ yes | ○ tiny models only | • Opus inf |
| Attention-side edits (§3) | ~ part | ○ tiny models only | • Fable r1, Opus r1 |
| Scaling and merging edits [25] | ✓ yes | ○ tiny models only | • Fable 5.2 rep |
| **Ship it** | | | |
| Export as a LoRA adapter [6] | ✓ yes | ● **real E4B** | — |

✓ a function and docs ~ building blocks only ✗ not in the library

● checked on the real E4B model ○ tested only on tiny random Gemma 4 models (Section 4)

**Runs of the companion study**

| | |
|---|---|
| **r1** | round 1: is the old text found at all? |
| **r2** | round 2: escape artefacts |
| **Opus** | Claude Opus 5.5 |
| **Fable** | Claude Fable 5.1; 5.2 where marked |
| **Astra** | GPT-6 Astra |
| **no edit** | Astra r2; Fable 5.2 inf and rep withdrew |
| **recipe** | our scripted difference-of-means ablation |
| **inf** | informed: read the earlier runs' records |
| **rep** | repair: began from a gradient-trained adapter |
| **round 1** | Opus, Fable, recipe |
| **round 2** | Opus, Fable, Astra, Opus inf, Fable 5.2 inf, Fable 5.2 rep |